

Question			Answer/Indicative content	Marks	Guidance
1			B	1	
			Total	1	
2	a		Area under graph = $0.5 \times 0.06 \times 1.8 = 0.054$ (Ns) $0.05 \times v = 0.054$, therefore $v = 1.1$ (ms ⁻¹)	C1 A1	
	b	i	Both forces shown in correct direction and arrows of same length. Zero.	B1	
		ii	(Conservation of momentum) $u_x = v_x + v_z$	B1	
		iii	(Conservation of kinetic energy) $u^2 = v^2$	C1	
		iii	$v_x^2 + v_z^2 = u^2$	C1	
		iii	Shows $v_x = 0$ by substitution	C1	
		iii	$v_z = u$ by substitution of $v_x = 0$	A1	
			Total	8	
3			A	1	
			Total	1	
			$T = 293$ K		
4		i	$\frac{3}{2} kT = \frac{1}{2} mv^2$	M1	
		i	$\frac{3}{2} \times 1.38 \times 10^{-23} \times 293 = \frac{1}{2} \times 4.7 \times 10^{-26} \times v^2$	C1	
		i	$v = 510$ (m s ⁻¹)	M1	
		i	1. Total vertical momentum after = 0 Total vertical momentum before = 0 (momentum is conserved)	A0	Note answer is 509.8 m s ⁻¹ to 4 s.f.
		ii		B1 B1	
		ii	$2. 4.7 \times 10^{-26} \times v \times \sin 88^\circ = 1.4 \times 10^{-24} \times \sin 45^\circ$	C1	
		ii	$v = 480$ (m s ⁻¹)	A1	Allow other correct methods.
			Total	7	
5		i	$250 \times 60 = 15000$ J	C1	

Question			Answer/Indicative content	Marks	Guidance
		i	energy = $\frac{15000}{0.65} = 2.3 \times 10^4$ (J)	A1	
		ii	drag force = $0.4 \times 6.02 = 14.4$ N	C1	
		ii	forward force = power / velocity = $250/6.0 = 41.7$ N	C1	
		ii	acceleration = $\frac{41.7 - 14.4}{85} = 0.32$ m s ⁻²	A1	
			Total	5	



EXAM PAPERS PRACTICE

Question			Answer/Indicative content	Marks	Guidance
6		i	N & W act on the same body / Newton's 3rd Law forces should act on different bodies	B1	Allow: 3rd law pair to W acts on (centre of) Moon 3rd law pair to N acts on surface of Moon
		i	N & W are different types (of force) / are not same type	B1	Allow: N is electromagnetic / electrostatic / electrical / contact W is gravitational. Allow: Paired forces should be of the same type Ignore a general statement of Newton's 2nd or 3rd law Examiner's Comments This question was designed to allow students to demonstrate their understanding of Newton's third law. Although many were able to quote the law accurately, few were able to apply it correctly. Only a minority of candidates scored any marks at all, usually for stating that the pair forces must be of the same type and N and W were not. It was particularly disappointing to see the large number of misconceptions offered such as 'Newton's laws only apply on Earth', ' N and W should be applied at the same point' and 'There is no gravity on the Moon'. Newton's third law continues to present a challenge to the majority of candidates. Do not allow 'acts on surface of Moon'
		ii	Equal to / same as W acting on (the centre of) the Moon	B1	Diagram is not sufficient for this mark Examiner's Comments This was a discriminating question with the more able candidates performing well. Many candidates attempting the question failed to read the word ' <i>location</i> ' and gave answers referring to just the magnitude and direction of the force.
Total				3	

Question			Answer/Indicative content	Marks	Guidance
7	a	i	$m = \frac{0.131}{6.02 \times 10^{23}}$ $m = 2.18 \times 10^{-25} \text{ (kg)}$	A1	Examiner's Comments Most candidates were able to score the mark for dividing the molar mass of xenon by Avogadro's number. A small number, however, obtained answers which were many orders of magnitude greater than the required value. Candidates are advised to ask themselves the simple question 'Is this a reasonable answer' before moving on to the next question. This check often suggests an error has occurred which may hopefully be found and corrected thus avoiding an unnecessary loss of marks.
		ii	mass of xenon ejected / s = $m_{Xe} = 2.2 \times 10^{-25} \times 9.5 \times 10^{18} (= 2.07 \times 10^{-6})$	C1	Possible ECF Allow: $5.2 \times 10^3 \Delta v = 2.07 \times 10^{-6} \times 3.2 \times 10^4$ $\Delta v = 1.3 \times 10^{-5}$ $a_S = 1.3 \times 10^{-5} \text{ (ms)}^{-2}$
		ii	$F_{Xe} = \left(m_{Xe} \frac{\Delta v}{\Delta t} \right) = 2.2 \times 10^{-25} \times 9.5 \times 10^{18} \times 3.2 \times 10^4 (= 0.06627)$ $a_S = \left(\frac{F_{Xe}}{m_s} \right) = \frac{2.2 \times 10^{-25} \times 9.5 \times 10^{18} \times 3.2 \times 10^4}{5.2 \times 10^3}$	C1	
		ii	$a_S = 1.3 \times 10^{-5} \text{ (m s}^{-2}\text{)}$	A1	
					Examiner's Comments There were three stages to this calculation: firstly, the determination of the mass of xenon ejected per second which most were able to complete successfully; secondly, the application of Newton's second law to determine the force on the xenon ions and hence, by Newton's third law, the force on the spacecraft; finally, the acceleration which could be calculated from Newton's second law. The majority of candidates followed this process through accurately and presented their working logically. The most common error, made by weaker candidates, was the use of the spacecraft's mass to determine the force on the xenon. This resulted in excessively high acceleration

Question			Answer/Indicative content	Marks	Guidance
		iii	Rate of change of momentum (of an object) is proportional to the resultant / net (external) force acting upon it. (AW) OR statement of law of Conservation of momentum in a closed system / no external forces	B1	 Momentum must be spelled correctly Allow: 'equal to' instead of 'proportional to' Allow: statement of Newton's 3rd Law provided it is clear the forces act on different bodies and opposite is spelled correctly Examiner's Comments The mark scheme was made particularly flexible here in order to reward candidates for accurately quoting either of Newton's second and third laws or a statement of conservation of linear momentum. This ensured that most candidates were in a position to score the mark. A significant number of candidates lost this mark by not making it clear that Newton's second law refers to the resultant force acting on the system. Enter text here.
		iv	Force (on spacecraft) is constant	B1	Not: Weight (of spacecraft) or 'it is lighter' Examiner's Comments Most candidates realised that the mass of the spacecraft would decrease as xenon is ejected and this would cause an increase in acceleration. Only a minority were aware that this conclusion required the force on the spacecraft to be constant and correctly drew attention to this fact. On the whole this question discriminated well across the range of candidates.
		iv	Mass (of spacecraft) decreases (as xenon is ejected)	M1	
		iv	Acceleration increases	A1	
	b	i	Area under graph in range 10.5 to 11.5 (Ns)	C1	
		i	Area under graph in range 10.8 to 11.2 (Ns)	C1	
			$\Delta v = \frac{\text{impulse}}{m} = \frac{\text{area}}{m}$		

Question			Answer/Indicative content	Marks	Guidance
		i	$= \frac{11.0}{180}$	C1	Possible FT for using their area / 180 Use of mass of spacecraft rather than satellite scores 1 out of last 2 marks.
		i	$= 6.1 \times 10^{-2} (ms^{-1})$	A1	Examiner's Comments Very many candidates realised that this question required them to calculate the impulse by finding the area under the curve. A wide variety of methods were seen but the poor layout made it difficult to follow the logic and identify specific errors. A significant number calculated this value by counting squares for the entire shape which inevitably consumed an excess of their time. The expected method of calculating the area of the trapezium from 0-12 ms at a height of 1000 N together with a rectangle from 4 to 9 ms of height 200 N using formulae, was not commonly seen. This method reduced the counting to a few small squares at the top of the graph. To reflect the time required to obtain a reasonably accurate area the examiners decided to award a maximum of 2 marks, dependent upon the accuracy of the value obtained. The candidate's area was then used to determine the change in velocity with error carried forward. This application resulted in good discrimination with about 75% scoring 2 or more marks. A small minority used the maximum force of 1300 N acting over a time of 12 ms; this approach received no credit. Allow: upper limit on time in range 3.0 to
		ii	From 0 to 3 (ms) acceleration increases linearly / uniformly / at constant rate / at a steady rate.	B1	3.5 ms Do not credit use of 'constantly' for this mark

Question			Answer/Indicative content	Marks	Guidance
		ii	(From 6.5 ms) onwards / later / at end the acceleration decreases	B1	Not 'decelerates' Examiner's Comments In this question it was common to see the phrase ' <i>rate of acceleration</i> ' used when clearly what was meant was ' <i>acceleration</i> '. Candidates should note that ' <i>rate of acceleration</i> ' is the 'jerk' experienced and this is the gradient of an acceleration against time graph. It is also inappropriate to talk of ' <i>acceleration slowing down</i> '. For the first marking point credit was only given if the candidate identified the increase in acceleration and stated that this occurred linearly between 0 and 3 ms. A small latitude was allowed on the upper limit of this range of times but a significant number lost this mark through inaccurate reading of the scale on the graph. The second mark was awarded for candidates correctly identifying the period from 6.5 ms as one in which the acceleration decreased without specifying the manner of the changes. Unfortunately many incorrectly associated the negative gradient with negative acceleration or deceleration rather than negative jerk and lost the second mark.
			Total	14	

Question			Answer/Indicative content	Marks	Guidance
8		i	$\rho = m/V = m/Av$; so $m = A\rho v$	C1	
		i	$7.5 \times 10^{-5} \times 1000 \times v = 0.070$	A1	
		i	giving $v = 0.93 \text{ (m s}^{-1}\text{)}$	A0	
		ii	$3.7 \text{ (m s}^{-1}\text{)}$	A1	Accept 3.72
		iii	$F = \Delta(mv)/\Delta t = 0.070 \times (3.72 - 0.93)$	C1	ecf (ii)
		iii	$F = 0.195 \text{ (N)}$	A1	accept 0.19 or 0.2(0)
		iv	arrow into the shower head perpendicular to its face.	B1	award mark for a reasonable attempt.
			Total	6	

Question			Answer/Indicative content	Marks	Guidance
9	a		A body will remain at rest or keep travelling at constant velocity unless acted upon by a resultant/net (external) force (AW)	B1	Allow 'speed in straight line' for velocity Allow 'uniform motion' Examiner's Comments Many candidates could not be awarded this mark because their answer did not emphasise the need for a resultant force to be acting. Allow act for the same time
	b	i	They have equal magnitude/ same size	B1	Allow have same line of action Examiner's Comments It was clear that candidates had been well prepared for this Newton's third law question. As a result many were able to score highly in both parts. The most common error made was concerning the location of the forces. Not act in different directions
		i	They are the same type / nature	B1	
		ii	Act in opposite directions	B1	Examiner's Comments It was clear that candidates had been well prepared for this Newton's third law question. As a result many were able to score highly in both parts. The most common error made was concerning the location of the forces.
		ii	Act on different bodies	B1	
	c	i	$\frac{dm}{dt} = \rho Av$ $= 1 \times 10^3 \times 3.3 \times 10^{-4} \times 25$ $(= 8.25 \text{ kg s}^{-1})$	B1	Examiner's Comments Well answered by almost all candidates.
		ii	Weight (of fireman) = $92g$ / $W = 92 \times 9.8(1)$ (= 903 N)	C1	Allow use of 8.3 leading to 170 N
		ii	Vertical component of water force = $8.25 \times 25 \sin 55$ (= 169 N)	M1	

Question			Answer/Indicative content	Marks	Guidance
		ii	Vertical component of contact force = 169 + 903 = 1100 N	A1	Note answer to 3 sf is 1070 N
					Note: a bald $\frac{92g}{\sin 55} = 1100$ is WP scores 0/3 Examiner's Comments Most candidates realised that the weight of the fireman was required in order to calculate the force exerted by the ground. However, only a minority were aware that the vertical component of the water force would create an additional component. It was rare to see a diagram used to support the answer. The most commonly seen error was the attempt to include a resolved component of the weight of water ejected.
			Total	9	
10			A	1	
			Total	1	
11	a		$mv - mu = F\Delta t$ ($u = 0$) / area under graph = Δp	C1	Allow 'impulse' for Δp
			$mv = \frac{0.010 + 0.040}{2} \times 150$	C1	Allow alternative methods for finding area.
			$mv = 3.75$ or $v = \frac{3.75}{0.16}$		
			$v = 23 \text{ (m s}^{-1}\text{)}$	A1	Note answer to 3 s.f. is 23.4 m s ⁻¹
	b		Curve upwards with decreasing gradient.	M1	
			Curve starts at a non-zero velocity at $t = 30$ ms.	A1	
			$F\Delta t = mv = mu$		Allow other correct methods
	c		$T = \frac{0.16 \times 23}{0.80}$	C1	Possible ECF from (a)
			$T = 4.6 \text{ (s)}$	A1	Note the answer is 4.7 (s) when 23.4 (m s ⁻¹) is used
			Total	7	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
12		i	$a = 4\pi^2 f^2 \times$	C1	condition for SHM
		i	so $k = (m4\pi^2 f^2) = 1.7 \times 10^{-27} \times 4 \times 9.87 \times 43.7 \times 1026$	B1	substitution
		i	$k = 292 \text{ (N m}^{-1}\text{)}$	A1	ecf if incorrect mass used
		ii	(N2 gives) $F_H = m_H a_H$ and $F_I = m_I a_I$	B1	allow total momentum = 0 at all times
		ii	(N3 gives) $F_H = F_I$ <i>can be implicit</i>	B1	SHM gives $v = 2\pi f x_{\text{max}}$
		ii	SHM gives $a \propto (-)x$	B1	so $m_H x_H = m_I x_I$
		ii	hence $x_H/x_I = a_H/a_I = m_I/m_H = 127$	B1	accept $127 = x_H/x_I \approx 10/0.08 = 125$
			Total		
			A	7	
13				1	Examiner's Comments This question was more challenging still. In this question, candidates often forgot that the impulse provided by the hockey stick is in the opposite direction to the momentum of the puck, again giving option A as the correct answer.
			Total	1	

Question		Answer/Indicative content	Marks	Guidance
14	a	${}^{238}_{92}\text{U} \rightarrow {}^{234}_{90}\text{Th} + \dots$ ${}^4_2\text{He} \text{ or } {}^4_2\alpha$	B1 B1	allow proton and / or nucleon number to the right of symbol allow γ -photon; zero for any other extra particle Examiner's Comments Most candidates made a good start to the paper writing a correct equation for the nuclear decay. allow 0.07×10^{-25} for α -particle mass
	b	$mv = (4.00 - 0.0665) \times 10^{-25} \times 2.40 \times 10^5$ $= 9.44 \times 10^{-20}$ $v = 9.44 \times 10^{-20} / 6.65 \times 10^{-27} = 1.42 \times 10^7$ $\text{k.e.} = \frac{1}{2} \times 6.65 \times 10^{-27} \times (1.42 \times 10^7)^2$ $= 6.70 \times 10^{-13} \text{ (J)}$ $6.70 \times 10^{-13} / 1.60 \times 10^{-13} = 4.19 \text{ (MeV)}$	C1 C1 A1 B1	max 3 if use 4.00 instead of 3.93 in momentum eq'n allow ratio of masses 234 and 4 or calculations using 234u and 4u allow $p^2/2m$ calculation for k.e. accept 4.0 to 4.2; ecf (calculated value of k.e. in J)/e N.B. the correct answer automatically gains all 4 marks Examiner's Comments One mark in this question was reserved for converting units from joule into mega electronvolt. This was the only mark awarded to half of the candidates. Few recognised this to be an isolated system, applying the conservation of momentum to solve the problem. Few appeared to realise that the mass of an alpha particle is given in the Data, Formulae, and Relationships Booklet, calculating it instead by summing the masses of neutrons and protons. The most common incorrect approach was to use the formula $E = mc^2$ or to equate the kinetic energies of the thorium nucleus and alpha particle.

Question			Answer/Indicative content	Marks	Guidance
	c		$\Delta A = 32 = 4n_{\alpha}$ so $n_{\alpha} = 8$ $\Delta Z = 10 = 2n_{\alpha} - n_{\beta}$ so $n_{\beta} = 6$ argument / reasoning given for both n_{α} and n_{β}	B1 B1 B1	allow 8 (decays), i.e no mention of α particles allow $10 - 16 = -6$; NOT $14 - 8 = 6$; must state $\beta(-)$ particles e.g. change in mass number caused by α decay, change in proton number combination of α and β Examiner's Comments A significant number had no idea where to start and left the page blank. Of the rest most managed to decide on 8 alpha particles. A minority worked initially with the proton number rather than the nucleon number incorrectly choosing 5. The explanations about the choice of 6 beta particles were often just restricted to equating the numbers correctly rather than giving any description of the transformation of neutrons into protons.
			Total	9	

Question			Answer/Indicative content	Marks	Guidance
15	a		$E_k = \frac{1}{2}mv^2$ <u>and</u> $p = mv$ (Correct manipulation leading to) $E_k = \frac{1}{2} \frac{p^2}{m}$	M1 A1	Allow: any subject Allow: $E_k = p^2/(2m)$
	b	i	From $t = 0$ to $t = 2.0$ s: a non-zero horizontal line From $t = 2.0$ to $t = 3.5$ s: line showing $v = 0$ From $t = 3.5$ to $t = 4.0$ s: non-zero horizontal line showing v is opposite in direction and magnitude <u>larger than</u> that of line drawn <u>at</u> $t = 0$ to $t = 2.0$. KE is constant.	B1 B1 B1	Judgement by eye
		ii	GPE increases linearly / proportional to t	B1 B1	Allow: 'at constant rate' for 'linear' Not: unqualified 'constantly' Examiner's Comments Nearly four fifths of candidates completed 20a well, especially if they clearly stated the equations for momentum and kinetic energy. Those that did not generally forgot that the question required an expression with 'p' and 'm' in it. $\frac{1}{2}pv$ was a common wrong answer. 20bi was answered well, with some candidates either slightly misreading the graph when the velocity became negative or not spotting that the line was steeper for the last section of the movement than it was in the first. Most candidates spotted that the KE was constant because the velocity was constant. Rather fewer candidates explained that the GPE increased <i>at a constant rate</i> .

Question			Answer/Indicative content	Marks	Guidance
		iii	$V^2 = 0.802 + 2 \times 9.81 \times 0.40$ $V = 2.9 \text{ (m s}^{-1}\text{)}$	C1 A1	Allow 1 mark for $(2 \times 9.81 \times 0.40)^{1/2} = 2.8 \text{ (m s}^{-1}\text{)}$ Examiner's Comments Many candidates selected the correct equation, although did not realise that the load was not at rest when it was released. The initial velocity was found from the graph on page 22 of the paper and was 0.80 ms^{-1} .
		iv	$F = 0.12 \times 2.9/0.025$ $F = 14 \text{ (N)}$	C1 A1	Possible ECF from (iii)1 Note: use of 2.8 m s^{-1} gives $F = 13(.44 \text{ N})$ Note: $1.4 \times 10n \text{ (N)}$ scores 1 mark Examiner's Comments Nearly three quarters of the candidates used the correct method for finding the average force acting on the load by considering the rate of change of momentum.
			Total	11	

Question			Answer/Indicative content	Marks	Guidance
16			Both forces act on the same object (AW) The types of forces are different / one force is gravitational and the other force is electrostatic	B1 B1	Allow: one force is gravitational (and the other is not) Examiner's Comments Many candidates incorrectly quoted Newtons' Third Law (N3L) or did not realise that a pair of N3L forces must be of the same type. The two forces in the question are acting on the same object, whereas N3L forces must act on different objects. Some candidates thought that the clothes were in motion, while the question states that the clothes are at rest. Normal Contact Force Normal contact force is essentially an electrostatic force between the two objects in contact.
			Total	2	

Question			Answer/Indicative content	Marks	Guidance
17		i	<u>KE</u> is conserved (as well as momentum)	B1	Allow: No <u>KE</u> lost
		ii	Attempt at conservation of momentum in x– or y– direction Correct expression of conservation of momentum in x– or y–direction / correct determination for velocity of Y of 55(3) m s^{–1} $p = 3.7 \times 10^{-24} \text{ (kg m s}^{-1}\text{)}$	C1 C1 A1	Allow confusion of sin and cos at this stage Allow attempt at conservation of KE Allow any subject e.g. $p \cos(25^\circ) + m \times 258 \cos(65^\circ) = m \times 610$ or $p \sin(25^\circ) = m \times 258 \sin(65^\circ)$ or $(p)^2 + (m \times 258)^2 = (m \times 610)^2$ or $\frac{1}{2}mv^2 + \frac{1}{2}m(258)^2 = \frac{1}{2}m(610)^2$ Answer is $3.67 \times 10^{-24} \text{ (kg m s}^{-1}\text{)}$ to 3 sf Examiner's Comments Most candidates correctly remembered that an elastic collision is one in which KE is conserved. In this series, it was acceptable to refer to 'no loss of KE'. Using the idea that the KE was conserved would have made calculating the velocity of particle Y straightforward. Most candidates preferred a conservation of momentum approach, which of course still works. Many candidates remembered that momentum is conserved in both the x– and y–directions independently and consideration of either was enough. Some candidates forgot that trigonometry was necessary to complete this step, or made errors with which of sine or cosine was required.
			Total	4	

Question			Answer/Indicative content	Marks	Guidance
18	a		$F = \frac{\Delta mv}{\Delta t}$ or $F\Delta t = \Delta mv$ or (resultant force) = rate of change of momentum area under graph = Δmv or Δp or change in momentum or impulse	B1 B1	Allow p instead of mv Allow: proportional for equals (rate of change of momentum) Examiner's Comments It was good to see that most candidates understood that Newton's second law of motion is more than the statement that $F=ma$. Many had successful attempts with some candidates missing that it is the rate of change of momentum, rather than the change of momentum that is required. About two-thirds of candidates also correctly indicated that the area under the graph represents the impulse or the change in momentum. In Question 20(b), some candidates assumed, incorrectly, that the maximum force multiplied by the time taken would give the change in momentum and so scored zero marks. Rather more simply divided the maximum force by the mass, which gave the right answer yet with incorrect physics. This approach also scored zero. In fact, more successful responses made it clear that the area of the triangle on the graph was the impulse and that that area gave a change in momentum of 900 Ns.

Question		Answer/Indicative content	Marks	Guidance
	b	area under graph = $0.5 \times 2.0 \times 900 = 900$ (N s) ($mU = 900$) $U = 13 \text{ (m s}^{-1}\text{)}$	C1 A1	Not: (initial force/mass) Examiner's Comments It was good to see that most candidates understood that Newton's second law of motion is more than the statement that $F=ma$. Many had successful attempts with some candidates missing that it is the rate of change of momentum, rather than the change of momentum that is required. About two-thirds of candidates also correctly indicated that the area under the graph represents the impulse or the change in momentum. In Question 20(b), some candidates assumed, incorrectly, that the maximum force multiplied by the time taken would give the change in momentum and so scored zero marks. Rather more simply divided the maximum force by the mass, which gave the right answer yet with incorrect physics. This approach also scored zero. In fact, more successful responses made it clear that the area of the triangle on the graph was the impulse and that that area gave a change in momentum of 900 Ns.

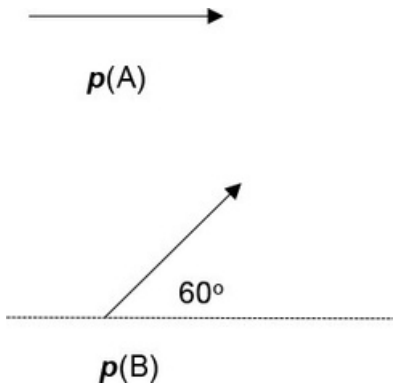
Question			Answer/Indicative content	Marks	Guidance
	c		The graph showing a (smooth) curve of continuously/always decreasing magnitude of gradient (with respect to time). Curve starts at $(0, U)$ and stops at $(2.0, 0)$	M1 A1	Note: curve must not be asymptotic at either end of the curve. Examiner's Comments <u>Successful candidates</u> spotted that the resultant force, the acceleration and hence the gradient of this speed- time graph decreased in magnitude with time. A constant gradient, ie a straight line between $(0, U)$ and $(2.0, 0)$, can only be achieved by a constant decelerating resultant force. This gives a curve that starts off $(0, U)$ with a steep negative gradient and finishes with a small negative gradient at $(2.0, 0)$.
			Total	6	

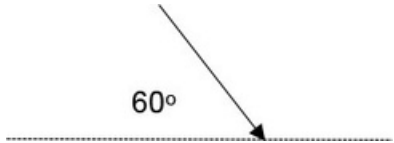
Question	Answer/Indicative content	Marks	Guidance
19	Level 3 (5–6 marks) Description and explanation of pattern changes and quantitatively explains link between de Broglie wavelength and potential difference. <i>There is a well-developed line of reasoning which is clear and logically structured. The information presented is relevant and substantiated.</i> Level 2 (3–4 marks) Clear description of how pattern changes and explanation of pattern changes and qualitatively explains link between de Broglie wavelength and potential difference or limited description of how pattern changes and quantitatively explains link between de Broglie wavelength and potential difference. <i>There is a line of reasoning presented with some structure. The information presented is in the most-part relevant and supported by some evidence.</i> Level 1 (1–2 marks) Limited description of how pattern changes and limited attempts to explain qualitatively the link between de Broglie wavelength and potential difference or qualitatively explains link between de Broglie wavelength and potential difference. <i>The information is basic and communicated in an unstructured way. The information is supported by limited evidence and the relationship to the evidence may not be clear.</i> 0 marks No response or no response worthy of credit.	B1 ×6	Indicative scientific points may include: Description of pattern changes • Rings become closer (not just smaller) • Rings become brighter Qualitative explanation of pattern changes in terms of de Broglie wavelength and potential difference • Electrons gain greater energy • Electrons have a greater speed • Electrons have a greater momentum • Implies smaller wavelength • Smaller wavelength means less diffraction • Shorter wavelength gives shorter path differences between areas of constructive and destructive interference Quantitative explanation of pattern changes in terms of de Broglie wavelength and potential difference • $eV = \frac{1}{2}mv^2$ • $p = mv$ • $v^2 \propto V$ or $p^2 \propto V$ • $\lambda = \frac{h}{p}$ or $\lambda \propto \frac{1}{v}$ • $\lambda = \frac{h}{\sqrt{2meV}}$ or $\lambda \propto \frac{1}{\sqrt{V}}$ <u>Examiner's Comments</u> This question tested an understanding of electron diffraction. Many candidates gave a good qualitative explanation of how the pattern would change. High achieving candidates clearly demonstrated how the de Broglie wavelength λ was related to the potential difference V by equating the energy eV to kinetic energy, then using the definition of momentum and the de Broglie wavelength. Some candidates confused speed v with potential difference V. Many candidates gave a good qualitative explanation. Many candidates did not state that the rings would become brighter.

Question			Answer/Indicative content	Marks	Guidance
					 AfL Candidates should be able to describe how to use light gates. In particular, candidates should be able to indicate the measurements that are needed to determine speed and acceleration. Candidates should state that the light gates should be connected to a timer or data-logger.  AfL When analysing experimental data, candidates should be able to determine appropriate graphs to plot which will give a straight line (if the given relationship is true). Candidates should also be able to describe how unknown quantities may be determined using the gradient and / or y-intercept.  Misconception There is some confusion between the equations to use for photoelectric effect and the equations to use when considering the de Broglie wavelength. For the de Broglie wavelength, a common misconception is to relate the energy to wavelength by the equation for the energy of a photon, $E = \frac{hc}{\lambda}$
			Total	6	

Question			Answer/Indicative content	Marks	Guidance
20			Level 3 (5–6 marks) Clear description of experiment and measurements and clear analysis. <i>There is a well-developed line of reasoning which is clear and logically structured. The information presented is relevant and substantiated.</i> Level 2 (3–4 marks) Some description of experiment and some measurements and some analysis. <i>There is a line of reasoning presented with some structure. The information presented is in the most-part relevant and supported by some evidence.</i> Level 1 (1–2 marks) Limited description of experiment or Limited measurements or Limited analysis <i>The information is basic and communicated in an unstructured way. The information is supported by limited evidence and the relationship to the evidence may not be clear.</i> 0 marks No response or no response worthy of credit.	B1×6	Indicative scientific points may include: Description • Release method • Ensure bob is not pushed • Repeat experiment for same H • Repeat for different H • Centre of mass of single bob and joined bob considered • Keep bob string taught Measurements • Measure heights h and H with ruler • Use centre of mass of bob or another suitable method • Use video camera to record motion • Use of datalogger and appropriate sensor to measure H and h • Measure mass with (top pan) balance Analysis • Construct a table of h and H • Plot graph of h against H • LoBF should pass through origin. • Determine gradient or calculate h/H repeatedly • gradient = $\left(\frac{M}{M+m}\right)^2$ (gradient must be consistent with the plot) • Masses substituted into above expression and checked against experimental gradient
			Total	6	

Question			Answer/Indicative content	Marks	Guidance
21		i	$(p =) 6.6 \times 10^{-26} \times 990$ or $6.5(3) \times 10^{-23}$ (kg m s ⁻¹) $(\Delta p =) 2 \times 6.6 \times 10^{-26} \times 990$ $\Delta p = 1.3 \times 10^{-22}$ (kg m s ⁻¹)	C1 A1	Ignore sign of answer
		ii	$990/[2 \times 0.46]$ (= 1080) $(F = \Delta p/\Delta t)$ $(F =) 1.3 \times 10^{-22} \times 1000$ $F = 1.3 \times 10^{-19}$ N	B1 C1 A1	Possible ECF from (b)(i) Note 1080 would give 1.4×10^{-19} (N)
		iii	Use of $p = F/A$ or pressure = (total) force / area Idea of multiplying by total number of atoms	B1 B1	Allow particles or molecules for atoms
			Total	7	
22	a		$(t =) 2 \times 1.3$ or 2.6 (s) $(x =) 68 \cos 11^\circ \times 2.6$ or 174 (m) horizontal distance = 174 – 90 horizontal distance = 84 (m)	C1 C1 A1	Note answer is 86 (m) if 1.32 s is used Note answer is 87 (m) if 1.3226... s is used Allow $1.3 \times 68 \cos 11^\circ$ for 1 mark Allow 3 or –3 m for 2 marks Allow KE is not conserved
	b	i	A collision in which kinetic energy is lost	B1	Not 'goes backwards'
		ii	Conservation of momentum Idea that velocity is to the right and velocity is very small / much smaller than 68 (m s ⁻¹)	B1 B1	
			Total	6	

Question			Answer/Indicative content	Marks	Guidance
23	a		(k.e. =) $E = 5.0 \times 10^6 \times 1.6 \times 10^{-19}$ $v = \sqrt{(2E/m)}$ or $= \sqrt{(2 \times 8.0 \times 10^{-13} / 6.6 \times 10^{-27})} = 1.6 \times 10^7 \text{ (ms}^{-1}\text{)}$ $p (= mv) = 6.6 \times 10^{-27} \times 1.6 \times 10^7$ giving $p = 1.1 \times 10^{-19} \text{ (kg m s}^{-1}\text{)}$	C1 C1 A1	$E = 8.0 \times 10^{-13} \text{ J}$ or ($E = p^2/2m$ so) $p = \sqrt{(2mE)}$ Note: A value of $v = 1.6 \times 10^7 \text{ (ms}^{-1}\text{)}$ automatically scores both C1 marks even if the calculation for E is not shown or $p (= \sqrt{(2mE)}) = \sqrt{(2 \times 6.6 \times 10^{-27} \times 8.0 \times 10^{-13})}$ giving $p = 1.0 \times 10^{-19} \text{ (kg m s}^{-1}\text{)}$ Full substitution of values must be shown and answer (if calculated) must be correct Examiner's Comments This question provided an excellent opportunity for candidates to produce immaculate responses: identify the physics involved, select and write down the correct formula, do the necessary conversion (MeV to J), rearrange the formula, substitute correctly and then write the final response in standard form to a correct number of significant figures. Some of the common errors were: • forgetting to convert 5.0 MeV into J • not showing a full substitution of values (which is necessary for a 'show that' question) • not calculating the response to more than 1 s.f. (which is necessary for a 'show that' question).
	b	i	Example (not to scale): 	B1 B1	horizontal arrow (judge by eye), in the direction shown arrow drawn at an angle of 60° to the horizontal (angle must be shown), in the direction shown

Question			Answer/Indicative content	Marks	Guidance
		ii	Example (not to scale):  (Can apply principle of) conservation of momentum (since no external forces are acting)	B1 B1	arrow drawn at an angle of 60° to the horizontal (angle must be shown), in the direction shown Examiner's Comments This was not an easy question but, even so, a good number of candidates did well. The marks were given for the direction (rather than for the magnitude) of the momentum vectors. Some of the common errors were: • forgetting to label relevant angles • not using arrows to show direction • drawing a vector triangle without any indication of which arrow was meant to be the final momentum.
			Total	7	